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How do digital technologies improve the physical health of higher education students? A systematic scoping review

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Abstract

Background The rapid evolution of digital technologies (DT) has reshaped health education, ushering in a new era of learning accessibility and interactivity within higher education (HE). This progress has sparked a growing interest in leveraging digital tools, ranging from mobile applications and wearable devices to online learning platforms and gamified interventions, to support various aspects of student physical health. However, the use of these technologies specifically to promote, monitor, and improve HE students' physical well-being remains a complex and rapidly expanding area that demands rigorous investigation. This systematic scoping review aims to explore and synthesize the current landscape of DT applications used to support the physical health of HE students, investigate their reported effects, analyze the practical implementation challenges, and assess future research directions.

Methods This study adheres to the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines. A comprehensive search was conducted across twelve databases, namely Web of Science, Scopus, PubMed, Medline, Embase, CINAHL, Cochrane Central Register of Controlled Trials, ACM Digital Library, IEEE Xplore, APA PsycInfo, ProQuest, and Google Scholar. A total of 16 articles were included in the study. Eligibility criteria were structured using the PCC (Population, Concept, Context) framework with explicit operational definitions for core terms.

Results The application of DT in HE student physical health is characterized by a deployment across multi-modal platforms and personalized, theory-driven interventions. The application landscape is dominated by integrated monitoring tools and mHealth apps, with a clear focus on addressing key physical health risks, promoting overall physical fitness, and enhancing curriculum delivery through flexible learning models. Efficacy is primarily demonstrated through improvements in engagement, fitness metrics, and behavior modification. However, limitations exist, including high rates of user discontinuation often due to technical unreliability, negative health consequences associated with excessive screen time, and risks related to generalizability due to an over-reliance on small, homogeneous convenience samples.

Conclusion Based on the included studies, DT shows potential as a scalable health management tool for providing HE students with personalized, continuous physical wellness support. However, the evidence base, derived from

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a limited number of heterogeneous studies, suggests that the core challenge for future development is to move beyond simple monitoring and achieve sustainable, long-term efficacy, which requires further robust investigation.

Keywords Digital technology, Higher education students, Physical health, Systematic scoping review

Introduction

The HE student population represents the future knowledge and innovation capital of society, and their health status holds profound significance for individual development and societal progress [1]. Physical health serves not only as a fundamental measure of personal quality of life but also as a core prerequisite for ensuring students' cognitive function, academic efficiency, and emotional stability [2]. In the university setting, good physical health can effectively mitigate anxiety related to academic stress, enhance concentration, and promote positive social interaction [3]. However, this cohort faces unique and increasingly complex health challenges. From a behavioral health perspective, HE students commonly face a dual health risk: the prevalence of physical inactivity and sedentary behavior [4]. Studies indicate that many university students fail to meet the minimum physical activity levels, increasing their health risk [5]. On the other hand, there are irregular routines and dietary habits [6]. Class schedules, social activities, and device use contribute to widespread poor sleep quality and sleep deprivation [7]. High-calorie, low-nutrient diets and stress-induced emotional eating are accelerating the spread of overweight and obesity issues across campuses [8]. The deterioration of student health not only leads to increased individual medical expenses and a higher risk of academic disruption but also poses challenges for HE institutions, requiring greater resource investment in intervention and support [9]. The early identification, effective intervention, and sustainable support of these physical health risks are crucial for mitigating their long-term negative consequences for individuals and society [10]. HE institutions and public health departments globally urgently need to develop and implement innovative and highly efficient health promotion strategies that can integrated into students' daily lives to systematically address key issues in maintaining HE students' physical health.

To address the health needs of HE students, traditional intervention and support models have long been dominant. These models primarily include face-to-face consultations at university health centers, group fitness classes offered at campus sports facilities, health education lectures conducted in dorms or cafeterias, and health information dissemination based on paper-based or basic digital tools [11]. Although these traditional models are grounded in professional clinical experience and educational theory and have played an initial role in raising awareness and disseminating information, they face multiple limitations when coping with the global and large-scale challenge of university student health. Firstly, there is the issue of resource accessibility and geographical equity. The number of professional health guides, e.g., nutritionists, sports coaches, and sleep specialists, often cannot meet the needs of the vast student population, and resources are mainly concentrated in large campuses or urban areas, leaving remote campuses and international students severely underserved [12]. Secondly, traditional models often lack real-time personalization and continuity. Traditional health assessments and tracking of intervention effectiveness largely rely on student recall and periodic surveys, making it difficult to capture students' behavioral fluctuations and physiological changes in real-time within unstructured environments [13]. This means interventions are often delayed and cannot be adjusted immediately based on dynamic individual states. Traditional models face challenges in terms of student engagement and adherence. Fixed times, locations, and formats struggle to attract modern university students, who have highly flexible schedules and a preference for digital interaction [14]. Many students may resist participating in in-person group programs due to time conflicts, social concerns, or insufficient attention to health. These limitations, particularly the lack of scalability, difficulty in achieving high-precision personalization, and challenges in overcoming geographic and psychological barriers, make traditional models ineffective in addressing the current complexity of physical health issues among HE students. This further highlights the urgent need for innovative, scalable, and highly efficient solutions.

Against this background, DT is emerging as a critical driver transforming HE student health service systems due to its unique advantages in speed and breadth. DT is a broad term encompassing all tools and platforms that leverage digital data, computing, and network communication to perform tasks, solve problems, or achieve goals [15]. In the health domain, this particularly refers to mobile Health (mHealth) applications [16, 17], wearable technology [18], telehealth platforms [19], AI-driven analytical systems, Virtual Reality (VR) and Augmented Reality (AR) interventions [20], and gamification platforms [21]. The core advantages of DT lie in its high accessibility, real-time capability, and personalization ability, allowing it to directly remedy the structural flaws of traditional health models [22]. Wearable technology, such as smartwatches and fitness trackers, non-invasively collects students' steps, heart rate variability, sleep stages, and oxygen levels in real-time [23]. Students can visually

see their sleep efficiency and deep sleep duration through an app, thereby increasing their attention to sleep hygiene [24]. mHealth applications utilize AI algorithms to provide highly customized health advice based on students' wearable data and self-reported preferences, for instance, adjusting calorie intake goals in real-time based on daily expenditure and diet logs [25]. By integrating gamification mechanisms into health interventions, such as reward points, badges, leaderboards, and virtual character progression, DT platforms have significantly boosted students' intrinsic motivation and long-term adherence [26]. Furthermore, telehealth platforms use video conferencing to allow students to have one-on-one online consultations and guidance with university sports coaches, registered dietitians, or physical therapists from any location, greatly overcoming geographical and time constraints [27]. VR and AR technology provide immersive exercise experiences, which are valuable for increasing the enjoyment of exercise and overcoming environmental limitations [28]. These multi-faceted applications of DT indicate that they are not merely simple alternatives to traditional health support but essential tools for achieving precise health management and large-scale personalized intervention, holding potential to change the HE student's physical health support paradigm.

Despite the widespread application of DT in HE students' physical health, existing research and reviews often suffer from fragmentation and a lack of systematic integration, which limits their utility for evidence-based practice. Some reviews exhibit the following limitations: Firstly, most reviews concentrate on a single technology, like only wearables or a single health outcome, e.g., only physical activity, and this point-wise approach fails to offer a comprehensive, cross-technology perspective to understand how DT impacts students' multiple health dimensions [29, 30]. Secondly, there is a lack of systematic exploration of practical application and implementation details. Existing reviews primarily focus on the quantifiable evaluation of intervention effectiveness (i.e., "is the technology effective?"). As in the research by Bi et al. [31], but pay insufficient attention to how the technology is effective, the implementation barriers, and the students' user experience. Furthermore, there is inadequate depth in discussing ethical, safety, and practical challenges. With the deeper integration of DT, ethical and practical challenges such as data privacy risks, algorithmic bias, health anxiety from information overload, and technological cost are becoming increasingly prominent.

To fill the gaps in the systematic and comprehensive scope of existing research, this study aims to conduct a thorough, systematic scoping review to investigate how DT helps HE students' physical health. Adhering strictly to the PRISMA-ScR guidelines [32], this study will systematically search, screen, and synthesize literature on the intervention and application of DT in the HE student physical health domain globally. The core objectives of this study are to comprehensively identify the DT types and currently DT applied to HE students' physical health; systematically analyze the reported effects and roles of these technologies in promoting specific health dimensions. Detail the specifics and procedures of DT's actual application and implementation in the HE environment, and clearly define the technical, ethical, and practical barriers present in the application of DT in this field, offering recommendations. This study will provide evidence-based guidance for administrators, health service departments, and technology developers at HE institutions globally, helping them select the most effective, cost-efficient, and student-appropriate DT solutions, and encouraging the development of safer, more user-friendly DT.

Based on the above background and objectives, this study proposes five core research questions:

1. What are the geographic and temporal trends in the application of DT to HE students' physical health, and what research methodologies are predominantly employed in this field?
2. What types of DTs are currently utilized to support HE students, and what are the primary physical health domains they target?
3. What are the common intervention forms of these DT applications, and what measurement indices are used to assess the reported effects of DT on HE students' physical health?
4. What barriers and challenges exist in the application of DT for promoting HE students' physical health?
5. What are the key future research directions for DT in promoting HE students' physical health?

Methods

This scoping review was conducted following the Joanna Briggs Institute (JBI) [33] methodology for scoping reviews and is reported in accordance with the PRISMA-ScR checklist [34] (see Appendix A).

Eligibility criteria

The eligibility criteria were defined using the PCC (Population, Concept, Context) framework to ensure clarity and focus.

Population

The population of interest was HE students, including undergraduate, postgraduate, and vocational training students, regardless of their field of study.

Concept

The core concept was the application of DT to improve, monitor, or support physical health outcomes. For this review, DT was operationally defined as any tool or platform using digital data, computing, or networking, including but not limited to mHealth apps, wearable devices, telehealth, VR/AR, gamification, and online platforms. Physical health outcomes were defined as measurable aspects related to bodily function and well-being, such as physical activity levels, sedentary behavior, cardiorespiratory fitness, body composition (e.g., overweight, obesity), sleep quality, and dietary habits. Studies focusing solely on mental health, academic performance, or institutional administration without a clear focus on physical health outcomes were excluded.

Context

The context for this review refers to the HE environment, encompassing both formal academic settings (e.g., physical education courses, campus facilities) and the informal daily life contexts of HE students where digital technologies are implemented to monitor or improve physical health.

Studies were included if they were original empirical research (e.g., randomized controlled trials, cohort studies, cross-sectional studies, qualitative studies) published in English with full text available. Non-academic reports, and studies solely focused on technology development, algorithm testing without field application, or those primarily exploring attitudes, perceptions, or conceptual frameworks without reporting intervention data or physical health outcomes were excluded. No date restrictions were applied. The final inclusion and exclusion criteria are summarized in Table 1.

Search strategy

A comprehensive search strategy was developed and executed on July 24, 2025. The search was conducted across twelve electronic databases and trial registries: Web of Science, Scopus, PubMed, Medline, Embase, CINAHL, Cochrane Central Register of Controlled Trials, ACM Digital Library, IEEE Xplore, APA PsycInfo, ProQuest, and Google Scholar, as well as ClinicalTrials.gov to identify ongoing or completed trials. The search strategy utilized a combination of keywords and subject headings related to three core concepts: (1) digital technology (e.g., "digital technology", "mHealth", "wearable device", "virtual reality", "gamification"), (2) higher education students (e.g., "higher education", "university student", "undergraduate"), and (3) physical health (e.g., "physical health", "physical activity", "sleep", "nutrition", "body weight"). The specific search queries for selected databases are provided in Table 2. The reference lists of included studies were also screened for additional relevant publications.

Study selection and data charting process

All records identified through database searches were exported to EndNote citation management software, where duplicates were removed. The study selection process involved multiple stages. Two reviewers (T.N. and T.L.) independently screened the titles and abstracts of all records against the eligibility criteria. The full texts of potentially relevant studies were then retrieved and assessed independently by the same two reviewers. Any disagreements at each stage were resolved through discussion or, if necessary, by consulting a third reviewer (Y.L.).

A standardized data charting form was developed and piloted on a sample of five included studies to ensure consistency. The term "data charting" is used consistently throughout this manuscript to describe the process of extracting data from included studies, in alignment with PRISMA-ScR guidelines. The final form captured the following information: author(s) and publication year; country of origin; study aims; research design/methodology; participant characteristics (target cohort, sample size); type of DT used; intervention form and duration; targeted physical health domains; measurement indices; key findings related to effects; reported barriers/challenges; and authors' conclusions regarding application

Table 1 Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
I1. Studies involving higher education students (undergraduate, postgraduate, vocational).	E1. Studies focusing on populations other than HE students.
I2. Studies investigating the application of digital technologies (as defined above) to support, monitor, or improve physical health outcomes (e.g., physical activity, sleep, nutrition, body composition).	E2. Studies focusing exclusively on mental health outcomes, academic performance, or institutional administration without a primary focus on physical health.
I3. Full text available and published in English.	E3. Full text not available or non-English publications.
	E4. Studies primarily exploring attitudes, perceptions, intentions, or conceptual frameworks regarding DT without reporting intervention data or physical health outcomes.
	E5. Studies solely focused on technology development, algorithm comparison, or laboratory-based testing without field application involving HE students.

Table 2 Selected databases and search queries

Database	Search query
Web of Science	(digital technology OR "DT" OR "mobile health" OR "mHealth" OR "wearable device" OR "fitness tracker" OR "smart watch" OR "health application" OR "health app" OR "telehealth" OR "telemedicine" OR "virtual reality" OR "VR" OR "augmented reality" OR "AR" OR "gamification" OR "eHealth") (All fields) AND ("higher education" OR "HE" OR "university student" OR "college student" OR "undergraduate" OR "postgraduate" OR "campus") (All fields) AND ("physical health" OR "physical activity" OR "exercise" OR "sleep" OR "nutrition" OR "diet" OR "body weight" OR "obesity" OR "well-being" OR "physical wellbeing") (All fields) AND (Document Types: Article or Proceeding Paper) AND (Languages: English)
Scopus	(TITLE-ABS-KEY ("digital technology" OR "DT" OR "mobile health" OR "mHealth" OR "wearable device" OR "fitness tracker" OR "smart watch" OR "health application" OR "health app" OR "telehealth" OR "telemedicine" OR "virtual reality" OR "VR" OR "augmented reality" OR "AR" OR "gamification" OR "eHealth")) AND (TITLE-ABS-KEY ("higher education" OR "HE" OR "university student" OR "college student" OR "undergraduate" OR "postgraduate" OR "campus")) AND (TITLE-ABS-KEY ("physical health" OR "physical activity" OR "exercise" OR "sleep" OR "nutrition" OR "diet" OR "body weight" OR "obesity" OR "well-being" OR "physical wellbeing")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp")) AND (LIMIT-TO (LANGUAGE, "English"))
PubMed	(((((digital technology) OR (DT) OR (mobile health) OR (mHealth) OR (wearable device) OR (fitness tracker) OR (smart watch) OR (health application) OR (health app) OR (telehealth) OR (telemedicine) OR (virtual reality) OR (VR) OR (augmented reality) OR (AR) OR (gamification) OR (eHealth)) AND ((higher education) OR (HE) OR (university student) OR (college student) OR (undergraduate) OR (postgraduate) OR (campus)) AND ((physical health) OR (physical activity) OR (exercise) OR (sleep) OR (nutrition) OR (diet) OR (body weight) OR (obesity) OR (well-being) OR (physical wellbeing)))) Filters: Full text
Medline	digital technology OR DT OR mobile health OR mHealth OR wearable device OR fitness tracker OR smart watch OR health application OR health app OR telehealth OR telemedicine OR virtual reality OR VR OR augmented reality OR AR OR gamification OR eHealth AND "higher education" OR HE OR university student OR college student OR undergraduate OR postgraduate OR campus AND physical health OR "physical activity" OR exercise OR sleep OR nutrition OR diet OR "body weight" OR obesity OR well-being OR "physical wellbeing"
Embase	'digital technology'/exp OR 'dt' OR 'mobile health' OR 'mhealth' OR 'wearable device'/exp OR 'fitness tracker' OR 'smart watch' OR 'health application' OR 'health app' OR 'telehealth'/exp OR 'telemedicine'/exp OR 'virtual reality'/exp OR 'vr' OR 'augmented reality'/exp OR 'ar' OR 'gamification' OR 'ehealth' AND 'higher education'/exp OR 'he' OR 'university student' OR 'college student' OR 'undergraduate' OR 'postgraduate' OR 'campus' AND 'physical health' OR 'physical activity'/exp OR 'exercise'/exp OR 'sleep'/exp OR 'nutrition'/exp OR 'diet'/exp OR 'body weight'/exp OR 'obesity'/exp OR 'well-being' OR 'physical wellbeing' AND [english]/lim AND [article]/lim OR [conference paper]/lim
CINAHL	((MH "digital technology") OR TI ("digital technology" OR "DT" OR "mobile health" OR "mHealth" OR "wearable device" OR "fitness tracker" OR "smart watch" OR "health application" OR "health app" OR "telehealth" OR "telemedicine" OR "VR" OR "AR" OR "gamification" OR "eHealth")) AND ((MH "university student") OR TI ("higher education" OR "HE" OR "university student" OR "college student" OR "undergraduate" OR "postgraduate" OR "campus")) AND ((MH "physical health") OR TI ("physical health" OR "physical activity" OR "exercise" OR "sleep" OR "nutrition" OR "diet" OR "body weight" OR "obesity" OR "well-being" OR "physical wellbeing"))
Cochrane	"digital technology" AND "higher education student" AND "physical health"
ACM Digital Library	("digital technology" OR DT OR "mobile health" OR "mHealth" OR "wearable device") AND ("university student" OR "higher education") AND ("physical health" OR "physical activity" OR sleep)
IEEE Xplore	("Full Text & Metadata": "digital technology" OR DT OR "mobile health" OR "mHealth" OR "wearable device" OR "VR" OR "AR" OR "gamification") AND ("Full Text & Metadata": "higher education" OR HE OR "university student" OR "college student") AND ("Full Text & Metadata": "physical health" OR "physical activity" OR exercise OR sleep OR nutrition) Filters Applied: Conferences Early Access Articles Journals
APA PsycInfo	"digital technology" AND "higher education" AND "physical health"
ProQuest	"digital technology" AND "higher education" AND "physical health"
Google Scholar	"digital technology" OR "DT" OR "mobile health" OR "mHealth" OR "wearable device" OR "VR" OR "AR" OR "gamification" AND "higher education" OR HE OR "university student" OR "college student" AND "physical health" OR "physical activity" OR exercise OR sleep OR nutrition OR diet OR "body weight"
ClinicalTrials.gov	Search Terms: university OR college OR "higher education" OR student Intervention/Treatment: "Mobile Health" OR mHealth OR "Digital Health" OR "Telemedicine" OR "Wearable Electronic Devices" OR "Health App" OR "Virtual Reality" OR "Gamification" Condition/Disease: "Health Behavior" OR "Physical Activity" OR "Exercise" OR "Sedentary Behavior" OR "Sleep Quality" OR "Dietary Habits" OR "Nutrition" OR "Obesity" OR "Body Composition" Study Type: Interventional Studies (Clinical Trials)

prospects and limitations. Data charting was performed independently by two reviewers (T.N. and T.L.). Inter-rater reliability, calculated using Cohen's kappa (κ), was substantial for both study selection (κ = 0.805) and data charting (κ = 0.773). Discrepancies in charted data were resolved through deliberation with the third reviewer (Y.L.).

Collating, summarizing, and reporting the results

The charted data were analyzed using a narrative synthesis approach to address the review questions. Quantitative data, such as the distribution of studies by year and country, and research methodologies, were summarized descriptively using figures and tables. Qualitative findings concerning DT applications, implementation, challenges,

and future directions were analyzed thematically. The entire author team reviewed and confirmed the interpretations and explanations presented in the manuscript to ensure accuracy.

Results

Search results

A systematic search across twelve electronic databases and two trial registries initially yielded 8,102 publications, with 3,873 duplicates removed to leave 4,837 unique records for title and abstract screening. This process excluded 4,795 irrelevant records, and the full texts of the remaining 42 articles were assessed for eligibility, resulting in the exclusion of 26 studies primarily due to misalignment with the PCC criteria, e.g., non-physical health focus, lack of field application involving higher education students. A final set of 16 studies was included in the synthesis (see Table 3), with the detailed selection process illustrated in the PRISMA flow diagram (Fig. 1).

Temporal and geographic distribution of studies

The included studies span a decade from 2015 to 2025, reflecting a growing research interest in the field. Publication frequency shows a notable upward trend: 1 study in 2015, 2 in 2020, 3 in 2021, 2 in 2022, 1 in 2023, 4 in 2024, and 3 in 2025 (Fig. 2). The surge in publications from 2023 to 2025 (10 studies, accounting for 62.5% of the total) coincides with the widespread adoption of consumer-grade digital technologies and the increased demand for scalable health interventions in higher education settings post-pandemic.

Geographic distribution of the research covers 10 countries/regions, with a concentration in technologically advanced and densely populated higher education systems (Fig. 3). China leads with 4 studies, followed by the USA with 3, while Spain and Malaysia each contributed 2 studies. Canada, Brazil, Finland, Lebanon, Taiwan, and one additional country each provided 1 study, reflecting regional priorities: Chinese research emphasizes large-scale digital monitoring and platform integration, U.S. studies focus on personalized mobile interventions, Spanish research explores gamification and digital teaching resources, and Malaysian studies center on digital feedback systems and online communities.

Research methodologies

Methodological diversity is evident across the 16 studies, with quantitative research dominating at 10 studies (62.5%), including randomized controlled trials (RCTs) with pretest-posttest designs, cross-sectional studies, and time-lagged designs. Analytical techniques range from basic inferential statistics (ANOVA, Chi-square tests) to advanced models (logistic regression, Confirmatory Factor Analyses) for validating complex relationships between technology use and health outcomes. Qualitative research accounts for 3 studies (18.75%), employing semi-structured interviews, focus groups, and thematic analysis to explore students' experiences, motivations, and barriers to technology use. Another 3 studies (18.75%) use mixed-methods research, combining quantitative outcome measures with qualitative data collection such as user feedback, expert interviews, and the Delphi method.

Digital technology applications and interventions

Types of digital technologies

The DTs used in the included studies are categorized into four functional modalities. mHealth applications and messaging form a foundational component, characterized by focused, accessible communication, including text messaging [35], HAPA-based mHealth interventions [36], and specialized tools like dietary assessment platforms [37]. Integrated tracking and monitoring tools emphasize objective data collection and immediate feedback, encompassing fitness apps, sports watches [38–41], wearable digital coaches [40], and digital monitoring systems that provide real-time performance data (e.g., heart rate, steps, activity completion). Experiential learning technologies include gamification-based programs [42], digital story creation [43], and online courses [44], which foster active engagement and flexible learning through reward mechanisms, creative reflection, and structured digital content. Comprehensive digital platforms and resources serve as centralized hubs, integrating multiple tools such as fitness trackers, interactive apps, VR, and learning management systems [45–47], while also encompassing remote teaching infrastructure that supports health-related education during crises [48].

Target cohorts and intervention forms

Target cohorts of the included studies vary in specificity and size, with sample sizes ranging from small (18 graduate students [49], 33 college students [35]) to large-scale (577 participants [47], 513 Chinese undergraduate students [46]). Specific subgroups include male college students with overweight and obesity [38], medical students [48], vocational training students [45], and student-athletes [41], while broader cohorts encompass healthy university students [36, 39] and general undergraduate/postgraduate populations [44, 49]. This diversity reflects the recognition that effective DT deployment requires tailoring content and delivery to specific subgroups within the higher education ecosystem, from high-risk populations to general wellness seekers across different educational stages (undergraduate, postgraduate, vocational training).

Intervention forms demonstrate sophisticated adoption of DT to facilitate behavioral change and educational

Table 3 Characteristics of the included studies

Author/Year/Country	Methods	Type of DTs	Target groups	Intervention form	Physical Health Outcomes / Target	Measurement index	Effect	Application prospects and limitations
Zhang et al. 2025 China [35]	Quantitative-randomized controlled trial (RCT) with pretest-posttest	mHealth Apps, Exercise Watches	65 Male college students with overweight and obesity	Online interventions carried out by the mHealth app	Overweight and obesity	Body composition test, maximum oxygen uptake test, maximum strength test, and physical fitness tests	Exercise intervention based on mHealth technology effectively improves the physical fitness of male college students with overweight and obesity. But online and offline groups were not significantly different.	mHealth is effective in managing college students' body fat percentage and improving physical health through exercise intervention. Objective data collected from wearable devices can ensure accurate and reliable measurement of physical activity, thereby minimizing the usual deviations associated with self-reported data. The relatively small sample size may limit the generalizability of the research results. The effectiveness of mHealth as an independent intervention measure for implementing health behavior interventions can be explored.
Tran et al. 2023 USA [36]	Quantitative-RCT	mHealth (ASA-24 dietary assessment tool)	28 college students	mHealth to Optimize Blood Pressure Improvement (MOBILE) Intervention	Diet and nutrition	Blood pressure (BP), complete anthropometric measurements (height, body mass), questionnaires	Non-nutrition-based mHealth program focusing on BP did not appear to improve dietary intake patterns.	The consumption location and the use of electronic devices seem to affect the dietary intake of college students. Measures for dietary intervention need to be developed to address appropriate eating habits, such as the influence of consumption location and electronic device usage, to optimize the consumption of college students. The research results are from a small sample of different ethnic college students at a single location. More research can be conducted to intervene in the eating habits of college students. Future research can focus on encouraging eating at home and limiting the use of electronic devices during meals.
Dillon et al. 2022 Canada [37]	Quantitative-RCT	mHealth (based Health Action Process Approach (HAPA))	61 healthy university students	Sedentary behavior-related planning + text messages intervention	Sedentary behavior	Time spent sitting, standing, walking, and stretching (using the Occupational Sitting and Physical Activity Questionnaire), frequency and duration of breaks from sitting (SIT-Q 7d questionnaire), action planning, coping planning, and action control constructs (SIT-Q 7d questionnaire), baseline characteristics (12-item modified Sedentary Behavior Questionnaire)	HAPA-based action planning and coping planning intervention, combined with tailored text messages, can promote reductions in student-related sitting time, while increasing time spent walking and stretching among university students.	Through a brief behavioral counseling course and subsequent daily text messages, the students' action plans, coping plans, and action control were successfully enhanced, thereby significantly reducing the sedentary time of this high-risk group of college students. Future studies need to conduct randomized controlled trials, including a larger sample size, device-based measurement of sedentary behavior, a longer follow-up period, health index assessment, and a more diverse sample.
Yan et al. 2015 USA [38]	Qualitative-formative participatory approach (focus groups)	mHealth (Text messaging)	33 college students	Developing a set of text messages promoting physical activity targeted to college students	Physical activity	Conduct framework analysis on the qualitative data obtained from the focus group discussions	Receiving Short Message Service (SMS) prompts that encouraged physical activity, and logging daily steps or minutes of exercise using SMS would function as a virtual exercise self-monitoring tool.	Accelerometer-based physical activity self-monitoring and mobile phone-based SMS feedback hold great promise for increasing physical activity. A small sample predominantly comprised of white women. They did not collect sufficient information to conclude the most effective recruitment strategy. Acceptability to a larger, more diverse body of college students is not known. Examines how physical activity-promoting messages can be delivered using various modalities and various channels. Investigates user experience with real-time assessment rather than simulated methods, and includes individuals from disadvantaged communities.

Table 3 (continued)

Author/Year/Country	Methods	Type of DTs	Target groups	Intervention form	Physical Health Outcomes / Target	Measurement index	Effect	Application prospects and limitations
Cabral et al. 2022 Brazil [39]	Quantitative-cross-sectional, observational study	Digital Teaching (Remote teaching)	123 medical students	Medical students to fill out a questionnaire	Sleep quality	A questionnaire about their perception of their sleep quality and academic performance, as well as about the presence of symptoms related to circadian rhythm disorders before the pandemic and during the period they underwent Remote Emergency Teaching	The attention span during remote classes and increased psychosocial stress related to the new form of adaptation.	Excessive screen time was associated with symptoms related to changes in sleep quality, with a decrease in academic performance, as well as increased levels of anxiety and stress. The physical symptoms were prevalent among medical students attending classes through emergency remote teaching. The importance of sleep hygiene, which is a regulating factor for academic performance and psychological fulfillment, considering the new forms of teaching implemented during the COVID-19 pandemic.
Mora-Gonzalez et al. 2020 Spain [40]	Quantitative-RCT	Gamification (Gamification-based program)	112 college students	15-week gamification-based program	Cardiorespiratory fitness (CRF)	Cardiorespiratory fitness assessment, attendance, number of weeks meeting physical activity recommendations, satisfaction assessment	Participants who met recommendations 100% of the time had a significant CRF improvement.	Gamification is a powerful tool for developing behavioral changes that lead to a healthy lifestyle in various contexts, among both young and adult populations. The convenience sampling meant that participants belonging to the IG were not randomly selected since they were already part of a specific student group of a subject intervention program. Gamification can lead to physical health benefits in youths. Educational institutions in the university context should target and promote PA throughout innovative strategies to lead young adults to improve their immediate and longer-term physical health.
Li et al. 2024 Malaysia [41]	Quantitative-RCT	Digital Monitoring (Digital monitoring and immediate feedback)	8,556 students	Students were explicitly informed that the physical activity they participated in would be digitally monitored, while the data generated from their running activities would be instantly fed back to the exercise data to the person through the smart sports education platform	Physical activity and fitness	Basic information, physical activity data, and physical fitness test results	The introduction of digital monitoring and immediate feedback technology significantly enhances undergraduates' participation in physical activity. The application of this technology not only enhanced students. motivation to participate but also provided personalized feedback, thus helping them to monitor and adjust their exercise behaviors more effectively.	The scope of technology application should be expanded from single exercise monitoring to include multiple dimensions such as exercise efficiency, recovery, nutritional intake, and psychological status to ensure that students' health and exercise effects are maximized. Students' heart rate, sleep quality, and activity intensity can be monitored through wearable devices combined with smart apps to provide personalized training advice and nutritional guidance. Through this all-round monitoring and instant feedback can not only help students avoid overtraining but also promote them to exercise healthily and scientifically.
Zheng 2024 China [42]	Quantitative	Digital Platforms	N/A	Biological data after physical exercise	Physical fitness	Physiological data processed both biologically and digitally	The integration of digital platforms significantly enhances the quality of physical education.	Aligning these practices with biotechnology-focused health education underscores the value of digital platforms in preparing students for a more integrated understanding of health, fitness, and biotechnology.
Kettunen et al. 2024 Finland [43]	Qualitative-interview (thematic analysis)	Wearable Devices (Digital coaches: Suunto 3 Fitness sport watch)	30 participants	Using the sport and wellness technology digital coach for 10–12 weeks	Physical fitness	The participants were asked to describe one single critical incident that occurred with the digital coach during the usage period in as much detail as possible.	As to the positive critical incidents, sentiments related to experiences of learning something new or being able to excel oneself were highlighted by the participants. On the contrary, sentiments related to a lack of trust in the accuracy of data were stressed mostly in the negative incidents.	Digital coaches have the potential to support physical activities and exercises. Participants only use the device's wrist-based heart rate measurement, which may affect the perception of data reliability. Since the reported events occurred at least at some point before data collection, there is a risk of recall and re-interpretation bias.

Table 3 (continued)

Author/Year/Country	Methods	Type of DTs	Target groups	Intervention form	Physical Health Outcomes / Target	Measurement index	Effect	Application prospects and limitations
Pickeral 2021 USA [44]	Mix-Quantitative (Pre- and post-assessments of push–up test), qualitative (face-to-face semi-structured interviews)	Digital Media (Digital story)	N/A	Participants were responsible for creating a digital story based on the physical activity they participated in over an 8-week study	Physical fitness	Pre- and post-assessments of push–up test, PACER test, height, and weight, face-to-face semi-structured interviews with five participants	There was a positive change based on physical assessment scores within the group that utilized the digital storytelling intervention.	Participants were not familiar with the concept of digital storytelling, but they had a positive experience using digital storytelling as an effort to increase their overall participation in and perception of physical activity.
Bardus et al. 2021 Lebanon [45]	Quantitative-questionnaire	Mobile Devices & Wearables (fitness apps, sports watch	230 students	Use of wearable devices, reasons for using wearable devices, reasons for stopping using wearables, use of health and fitness apps, reasons for using such apps, and reasons for stopping using apps	Physical fitness	A questionnaire -LimeSurvey Version 3.14.8 + 180,829	About a third of the sample, representing about 41% of the entire population of student-athletes, this segment of the population represents the 'innovators' or 'early adopters' of technology in the niche market of wearable devices	Some student-athletes use wearable devices and health apps to track their activity and monitor their health behaviors. These devices are used by a relatively small number of people, which constitutes a niche market. Longitudinal studies are needed to understand how student-athletes use these technologies in their daily routines and what could prompt continued use so that these devices could be effectively used to improve student-athletes' fitness and performance.
Marín-Marín et al. 2020 Spain [46]	Quantitative-pretest-posttest	Digital Resources (mobile application, learning management system platform)	177 vocational training students	A mobile application and a learning management system platform were used to host the content in a complementary manner. The teacher was solely responsible for guiding the students and answering their questions regarding the use of the mobile application, accessing or understanding the digital content on the platform. Students could use their own devices to access their resources	Diet habits	Questionnaire-ECHAES	The use of digital resources to educate students about diet habits and active lifestyles is effective in comparison with a traditional teaching method in the students of the Higher Technician in Dietetics.	The innovative teaching method that teaches dietary habits and an active lifestyle can better assist students in mastering knowledge and learning effectively. The implementation of teaching methods based on digital resources can facilitate the acquisition of dietary education content. This teaching method can promote the training of other teachers. In the future, new e-learning projects should be promoted so that students can use them at home to ensure that they can learn and practice healthy habits and be provided with a tracking system to obtain necessary feedback.
Chao et al. 2021 Taiwan [47]	Mixed-methods (Quantitative: questionnaire; Qualitative: focus group interviews)	Online Courses (Online dance course)	290 participants	A 10-week online dance course	Physical fitness	Physical Activity Class Satisfaction Questionnaire (PACSQ)	Applying blended learning instead of the traditional teaching method could indeed improve the Taiwanese students' learning effects regarding their dance skills and their satisfaction with this physical activity course in the aspects of cognitive development, fun and enjoyment, diversionary experiences, and relaxation.	This study covered dance instruction only and thus has its limitations. This study suggests that when teaching Taiwanese students and foreign students in dance courses, teachers can consider applying BL to improve their learning effect and satisfaction. If all the students are from overseas, faculty may also consider including flipped learning as an auxiliary for blended learning, as it might lead to better teaching results and quality.

Table 3 (continued)

Author/Year/Country	Methods	Type of DTs	Target groups	Intervention form	Physical Health Outcomes / Target	Measurement index	Effect	Application prospects and limitations
Congmeng and Mazlan 2025 Malaysia [48]	Qualitative-semi-structured interviews (Three-step thematic analysis)	Online communities	18 graduate students	It investigates how these factors facilitate or hinder skill development by examining teaching techniques and interventions in graduate education	Physical fitness	Elicit rich narratives regarding their adoption of educational technology, engagement with online communities, experiences with self-paced learning, and the perceived impact on their health promotion skills	Interactive educational platforms increase students' participation and make it easier for them to acquire health-related information. By providing students with a wide range of health information, educational technology can help overcome geographical barriers.	When applying these research results to other disciplines or degrees, the limitations and key points of the research for graduate students in physical education should be taken into account. Although this is sufficient for qualitative research, the sample size may limit the application scope of the research results. Future research should involve more disciplines and institutions to enhance external validity. Self-report bias is another consideration for qualitative research. This study is limited to graduate students, so it is difficult to apply to undergraduate students. Future research could explore how educational technology, online communities, and self-directed learning affect students' health promotion skills, incorporating gradual behavioral and attitudinal changes into this study.
Cui et al. 2024 China [49]	Quantitative-questionnaire	Fitness trackers, interactive apps, VR, and online platforms	513 Chinese undergraduate students	The time-lagged design to investigate the impact of technology use in physical education classes on fitness outcomes, mediated by student engagement and moderated by personal attributes. The research unfolded across three waves spanning over 6 months	Physical fitness	Modified Fennema-Sherman attitude scales, questionnaires were utilized as the primary data collection tool, featuring items designed to gauge perceptions and behaviors across several dimensions related to technology use, student engagement, fitness outcomes, and personal attributes in physical education settings	The incorporation of technology in physical education leads to improved physical health metrics among students. The role of student engagement as a mediator adds depth to our understanding.	The sample was limited to Chinese undergraduate students, which restricts the generalizability of the findings to other cultural or age demographics. In the reliance on self-reported measures for technology use, student engagement, and fitness outcomes. the relatively short time frame of this study limits the ability to make conclusive statements about the long-term effects of technology use on fitness and engagement. The study also did not differentiate between various types of technology, which may have different impacts on student engagement and fitness outcomes. Future research should include more diverse populations from different regions and educational levels, incorporating objective measures of fitness and engagement, extending the study's duration, and exploring a wider range of moderating variables.
Li et al. 2025 China [50]	Mix (Quantitative: questionnaire; Qualitative: expert interviews, and Delphi method)	Digital platform for aerobic dance	577 participants	Over 12 weeks and included: Survey Administration, expert interviews, classroom observations, digital platform usage Logs	Physical fitness / Skill development	Distribution of questionnaires to students and instructors to collect data on engagement and learning outcomes. Conducted with specialists to gather qualitative insights. Instructors monitored student participation during aerobic dance sessions to evaluate practical engagement. Data were collected on student interaction with the platform, including time spent, module completion, and feedback sessions	The DT-Based Aerobic Dance Teaching Model incorporates short, on-demand videos and interactive multimedia that allow students to engage with learning materials both synchronously and asynchronously. This flexible access to content ensures that students can practice dance movements outside of the traditional classroom setting, extending learning opportunities and reinforcing skills through repetition and self-paced study.	The DT-Based Aerobic Dance Teaching Model effectively enhances student engagement, learning flexibility, and performance in aerobic dance education. This model modernizes traditional teaching by integrating digital platforms, real-time feedback, and interactive learning, making it scalable and adaptable for university settings. For university administrators, investing in IT infrastructure and digital platforms enhances resource efficiency and reduces classroom dependency. For instructors, the model supports both structured and flexible teaching, allowing for better monitoring of student progress and the adoption of adaptive teaching methods. Future research should explore its effectiveness in diverse contexts to confirm its scalability and adaptability.

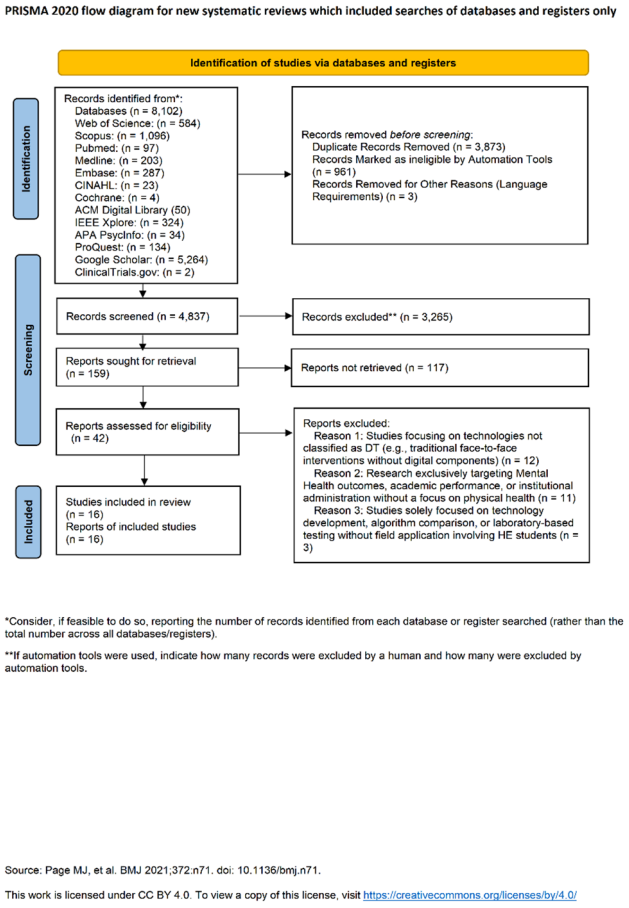


Fig. 1 PRISMA flowchart

delivery, falling into three core strategies. Behavioral nudge and feedback systems rely on targeted, personalized reinforcement, including sedentary behavior-related planning with text message interventions [36], physical activity-promoting text messages [35], digital monitoring with immediate feedback [39], and digital coach interventions [40], with durations ranging from 8 to 15 weeks. Structured educational integration embeds DT into formal coursework, such as 10-week online dance courses [44], mobile applications combined with learning management system platforms [45], 15-week gamification-based programs [42], and technology-integrated physical education classes with long-term (6-month) follow-up [46]. Holistic program delivery and assessment include questionnaire-based assessments of wearable device use [41], multi-faceted data collection (surveys, expert interviews, classroom observations, digital platform logs) over 12 weeks [47], and questionnaire assessments of remote teaching impacts [48], serving both intervention and baseline measurement purposes.

Targeted health domains and measurement

Physical health domains

The targeted physical health domains show a concentrated yet diversified focus, with physical fitness as the dominant category (7 studies) addressed through interventions like mHealth-based exercise programs [38, 39], online courses [41], and digital platform-integrated physical education [46, 47]. Other domains include diet and nutrition (2 studies) focusing on dietary intake patterns [37] and habit formation [45], sedentary behavior (1 study) targeted through behavioral planning interventions [36], cardiorespiratory fitness (1 study) improved via gamification [35], overweight and obesity (1 study) managed through exercise-focused mHealth interventions [38], sleep quality (1 study) assessed for impacts of remote teaching [48], aerobic dance performance (1 study) enhanced through digital teaching models [47], and physical activity (1 study) promoted via text message prompts [35]. This distribution underscores the field's

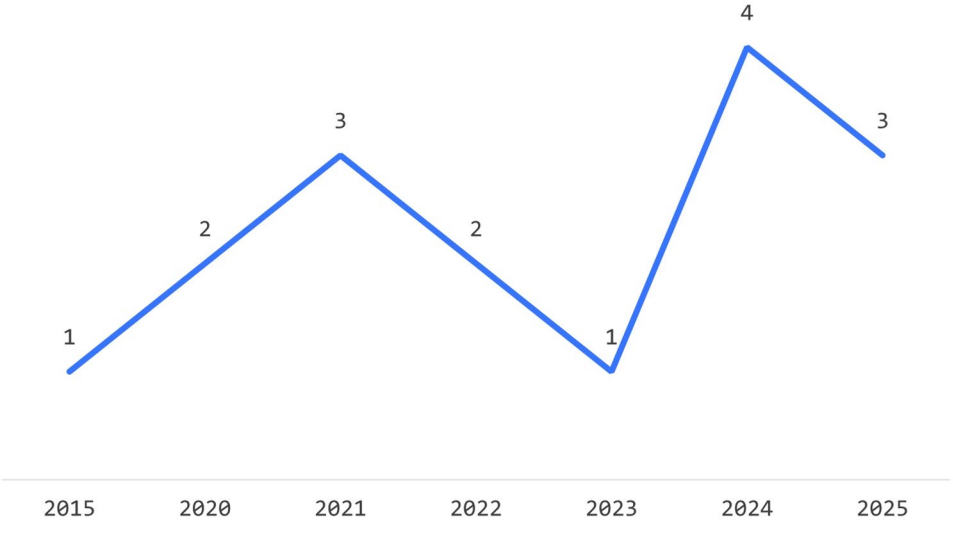


Fig. 2 Annual number of publications

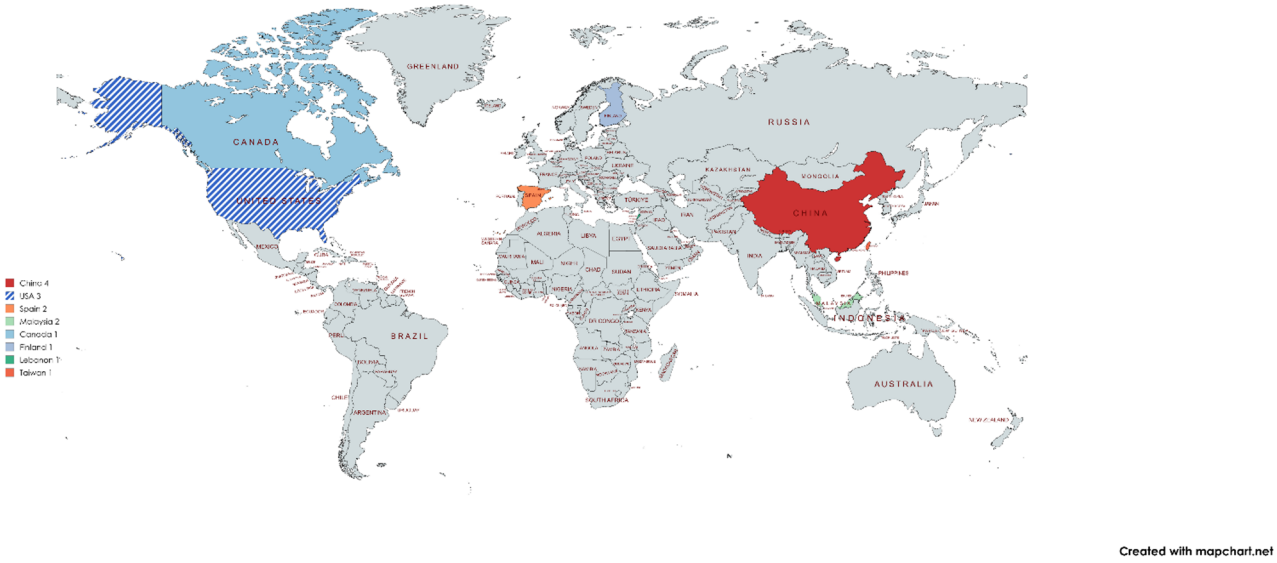


Fig. 3 Countries of publication

focus on both general fitness enhancement and targeted modification of key lifestyle behaviors (sedentary behavior, diet, sleep) that are critical to student health in higher education settings.

Measurement indices

Measurement indices are grouped into three core categories, reflecting a multi-modal approach to validating intervention impact. Objective physiological and fitness tests include body composition tests, maximum oxygen uptake tests, strength tests [38], blood pressure and anthropometric measurements [37], push-up tests, PACER tests [43], cardiorespiratory fitness assessments [42], and biologically processed physiological data [50], providing verifiable evidence of physical changes. Subjective behavioral and perception assessments utilize questionnaires on diet habits [37, 45], sleep quality [48], sedentary behavior [36], and attitude scales [46], as well as semi-structured interviews, focus groups [35, 49], and critical incident techniques [40] to capture psychological and behavioral context. Digital engagement and usage data include attendance records [42], physical activity recommendation adherence weeks [39], and digital platform usage logs (time spent, module completion, feedback sessions) [47], ensuring that reported effects are attributable to sustained engagement with the technology rather than short-term novelty.

Reported effects and implementation challenges

Reported effects of DT interventions

Reported effects of DT interventions include both positive outcomes and null/limited results. Positive effects include improvements in physical fitness, e.g., enhanced strength, cardiorespiratory capacity [38, 39, 42], reduced sedentary time and increased walking/stretching duration [36], enhanced physical activity participation and motivation [35, 39, 43], improved mastery of dietary education content [45], increased self-monitoring of physical activity [39, 40], and improved skill learning and course satisfaction in digital-integrated physical education [44, 47]. Null/limited effects include non-nutrition-based mHealth programs showing no improvement in dietary intake patterns [37] and no significant difference in physical fitness outcomes between online and offline mHealth intervention groups [38]. These findings highlight the importance of aligning intervention design with targeted health domains to ensure effectiveness, as generalized approaches may fail to address specific behavioral or physiological needs.

Barriers and challenges in implementation

Barriers and challenges to DT implementation are multifaceted. Technical unreliability and data accuracy concerns erode user trust and lead to discontinuation [40, 41]. High rates of disengagement are also reported due to loss of interest in novelty-driven tools or technical faults in wearable devices and apps [41]. Negative health consequences associated with excessive screen time, such as reduced sleep quality, increased anxiety, and psychosocial stress, emerge as a paradoxical risk of digital health interventions, particularly in remote teaching contexts [48]. Digital divide issues, including connectivity problems, inadequate technological infrastructure, and unequal access to devices and data plans, undermine equitable access to digital health resources, especially in developing regions. Ethical challenges include data privacy risks, algorithmic bias, and reliance on self-reported data that may suffer from recall bias and social desirability effects

[37, 46], compromising the validity of research findings. Additionally, the over-reliance on small, homogeneous convenience samples in many studies limits the generalizability of findings to broader higher education student populations, while the lack of long-term follow-up prevents assessment of sustained behavior change and intervention durability.

Discussion

Synthesis of core findings

The synthesis of 16 studies provides a comprehensive overview of the current landscape of DT applications in promoting HE students' physical health, addressing the core research questions posed in this scoping review. Geographically and temporally, the field shows accelerated growth since 2023, driven by the mass adoption of consumer-grade DT and institutional demand for scalable wellness solutions, with research concentrated in technologically advanced regions with distinct regional priorities, such as large-scale monitoring in China, personalized mobile interventions in the USA, gamification and digital teaching in Spain, and digital feedback systems in Malaysia. Methodologically, the dominance of quantitative research reflects a focus on measurable efficacy, while the inclusion of qualitative and mixed-methods studies underscores the need to complement statistical validation with insights into user experience, barriers, and adherence.

Technologically, the landscape is characterized by multi-modal functionality, with four core categories: mHealth apps/messaging, integrated tracking/monitoring, experiential learning technologies, and comprehensive digital platforms, addressing a range of health domains from general physical fitness to targeted behaviors like sedentary behavior, diet, and sleep. Target cohorts vary widely, from small, specialized subgroups to large-scale general student populations, reflecting the adaptability of DT to different health needs and educational stages. Intervention forms range from short-term behavioral nudges to long-term curriculum integration, with measurement indices combining objective physiological data, subjective assessments, and digital engagement metrics to provide a holistic view of impact. While positive effects are consistently reported across fitness, behavior change, and educational outcomes, null results in specific domains highlight the importance of targeted design, and multiple barriers, technical, ethical, and practical, pose challenges to sustainable implementation. Our findings align with broader evidence in young adult populations suggesting that mHealth apps are effective for short-term behavior change, yet we extend this by identifying specific challenges unique to the HE environment, such as the conflict between screen-based learning and sleep hygiene [51]. Although digital health is on the rise across the entire population, its socialization and Gamification characteristics are particularly prominent among HE students, which is highly consistent with this group's preference for digital interaction [52].

Interpretation of intervention effectiveness

The synthesis of the included studies suggests that DT holds potential as a multi-modal resource for enhancing the physical well-being of HE students, with reported efficacy across behavioral modification, physiological improvement, and educational enhancement. The core strength of DT appears to lie in its capacity for personalized, theory-driven intervention. Studies utilizing models like the HAPA [36] indicate that micro-interventions, such as tailored text messages and planning prompts, may promote reductions in high-risk sedentary behavior and increase physical activity, a potentially key finding given that sedentary time is a major academic and health challenge. This potential efficacy may extend to targeted physiological outcomes; for instance, mHealth-based exercise programs are suggested to be effective in improving physical fitness and body composition in high-risk groups [38]. The use of digital monitoring and immediate feedback is reported to enhance undergraduates' participation and motivation by providing instant data on performance [39], a mechanism that potentially leverages the digital native tendencies of this cohort. DT may also transform pedagogy; models like the DT-based aerobic dance teaching model [47], and the use of digital resources for dietary education [45] might modernize curricula by offering flexible, asynchronous access to high-quality content. This blended learning approach is reported to not only improve skills and learning satisfaction but also enhance student engagement, which several studies identify as a potentially critical mediator between technology incorporation and improved physical health metrics [46]. However, the findings also indicate that effectiveness is not universal; for example, some general mHealth programs reportedly fail to improve specific behaviors like dietary intake patterns [37], emphasizing that interventions should be behaviorally specific and well-aligned with the targeted health domain.

Technical, ethical, and practical barriers

Despite the positive outcomes reported, the broad and sustainable implementation of DT in HE student health appears constrained by a series of technical, ethical, and practical hurdles that necessitate a systemic institutional response. Technical and data integrity issues may pose a significant threat to long-term trust and adherence. While the shift to objective data from wearable devices is often cited as an advantage, data quality remains a concern, as users frequently report a lack of trust in data accuracy due to sensor limitations or technical faults,

which often lead to discontinuation of use [40, 41]. This potential unreliability can be compounded by the ethical challenge of relying on self-reported data, which may suffer from recall bias and social desirability effects, potentially compromising the validity of research findings [37, 46]. There are also fundamental challenges, particularly in developing nations, relating to connectivity problems, lack of adequate technological infrastructure, and student disparities in access to devices and data plans, which could create a digital divide that undermines equitable access to digital health resources. Practical and behavioral integration barriers may stem from the very nature of digital life. Studies highlight that excessive screen time, especially during emergency remote teaching, is associated with negative health outcomes, including a decrease in sleep quality, increased anxiety, and psychosocial stress [48]. This suggests DT itself could be a risk factor if not managed through rigorous digital literacy and hygiene interventions. The findings that wearable adoption often remains a niche market among certain student groups, and that a significant proportion stops using apps due to loss of interest [41], indicate a need for interventions to move beyond novelty and integrate into the core routines of student life. Ethical and systemic generalizability challenges are tied to research scope. The reliance on relatively small, convenience samples from specific regions potentially limits the external validity and cultural adaptability of findings to global and diverse student populations. As DT interventions become deeply personal, the need for stringent governance of digital media, data security, and privacy protection becomes paramount. HE institutions might need to focus not only on the technical implementation but also ensure that their digital health platforms are aligned with robust ethical frameworks to safeguard sensitive student health information.

Future research directions

The trajectory of future research could address the identified gaps by potentially prioritizing rigor, generalizability, and sustainable implementation strategies. The field might transition from exploratory proof-of-concept studies to a focus on holistic, system-level validation. A potential priority is to conduct large-scale RCTs with significantly longer follow-up periods that incorporate objective measures, e.g., utilizing accelerometer-based data rather than self-report, to more conclusively determine the long-term effectiveness and maintenance of health behaviors. These studies could systematically include more diverse populations from different regions and educational levels to enhance external validity and establish culturally relevant benchmarks. Future innovation might focus on multi-dimensional monitoring and integration. This could involve expanding the DT scope from simple activity tracking to a more holistic framework that fuses physiological signals with behavioral metrics, e.g., nutritional intake, recovery status via smart apps and wearables. This deeper integration may be valuable for providing personalized, context-aware training advice considered necessary to support student health outcomes. Research could also investigate the causal pathways of engagement, exploring how factors like gamification, online communities, and institutional support mediate the relationship between technology use and sustained physical activity. A dedicated research stream might be needed to address sustainability and systemic alignment. This could include developing frameworks to assess and improve the digital competence and health literacy of students and faculty alike, ensuring they can effectively utilize these tools. Research might explore innovative pedagogical approaches, like flipped learning or blended learning, that strategically integrate DT while aiming to minimize the negative effects of excessive screen time on sleep and stress. For university administrators, future research could provide evidence-based guidance on investing in IT infrastructure and resource-efficient digital platforms that aim to enhance student care while considering dependency on traditional delivery models.

Limitations

The conclusions drawn from this systematic scoping review, while providing a comprehensive overview of the current landscape of DT in improving the physical health of HE students, are subject to inherent limitations. Firstly, the included studies, even when published recently, may not fully capture the newest generations of devices, sophisticated platforms, or emergent ethical risks, posing some challenge to the timeliness of the synthesized findings. Secondly, this review was strictly limited to empirical studies focusing explicitly on physical health outcomes. This necessary methodological rigor means the review intentionally excluded potentially valuable insights from non-empirical research. Thirdly, the included studies exhibited wide variability in sample size, intervention duration, technology used, and outcome metrics. While the analysis addressed this heterogeneity through descriptive categorization, the inherent differences prevent the execution of a formal meta-analysis and limit the ability to draw definitive conclusions about comparative effectiveness. The future research directions identified in this review are fundamentally significant because they provide a strategic roadmap to overcome the limitations of the existing literature base. By advocating for larger, long-term RCTs, objective outcome measures, and culturally diverse samples, these directions guide the field toward generating the robust evidence required to translate promising DT applications into effective, institutional-grade health policies.

Conclusion

This systematic scoping review, based on the synthesis of 16 studies, suggests that DT hold potential as a resource for supporting the physical health of HE students. The reviewed literature indicates that DT applications may contribute to engagement and behavior change through features like continuous monitoring, immediate feedback, and personalized interventions, with some studies reporting associated gains in physical fitness metrics and educational satisfaction. However, the evidence base also highlights significant challenges. The sustainable large-scale deployment of DT appears critically challenged by technical issues such as data unreliability and high rates of user discontinuation. Paradoxically, the interventions themselves may carry behavioral risks, such as negative health consequences associated with excessive screen time. Furthermore, the generalizability of findings is limited by the over-reliance on small, homogeneous samples and a lack of long-term follow-up in the existing research. Therefore, while DT shows promise, the current evidence does not strongly support its unequivocal efficacy. Future progress in this field hinges on overcoming these limitations through more robust, long-term, and diverse studies that can provide clearer evidence of effectiveness and guide the development of sustainable, ethically sound implementation strategies. Only through such focused efforts can the potential of DT as a tool for HE student wellness be more reliably assessed and realized.

Appendix 1

Preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews (PRISMA-ScR) checklist

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
TITLE			
Title	1	Identify the report as a scoping review.	Cover page
ABSTRACT			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	2-3
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	4
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	4-6
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	4-6
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	4-6
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	4-6
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	4-6
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	6-7
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	6-7
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	6-7
RESULTS			
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	7

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	7
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	7
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	8-19
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	20-25
DISCUSSION			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	25-27
Limitations	20	Discuss the limitations of the scoping review process.	27-28
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	28
FUNDING			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	Not applicable

Note: JBI = Joanna Briggs Institute; PRISMA-ScR = Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews

* Where sources of evidence (see second footnote) are compiled from, such as bibliographic databases, social media platforms, and Web sites

† A more inclusive/heterogeneous term used to account for the different types of evidence or data sources (e.g., quantitative and/or qualitative research, expert opinion, and policy documents) that may be eligible in a scoping review as opposed to only studies. This is not to be confused with information sources (see first footnote)

‡ The frameworks by Arksey and O'Malley (6) and Levac and colleagues (7) and the JBI guidance (4, 5) refer to the process of data extraction in a scoping review as data charting

§ The process of systematically examining research evidence to assess its validity, results, and relevance before using it to inform a decision. This term is used for items 12 and 19 instead of "risk of bias" (which is more applicable to systematic reviews of interventions) to include and acknowledge the various sources of evidence that may be used in a scoping review (e.g., quantitative and/or qualitative research, expert opinion, and policy document)

Authors' contributions
T.N. and T.L. proposed and designed this study. Y.L. helped with data analyses and resolved disagreements in the review process. T.N. wrote the manuscript with the assistance of T.L. P.P. supervised and provided resources for this work. All authors discussed the results and contributed to the final manuscript.

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Data availability
The author confirms that all data generated or analyzed during this study are included in this published article. Furthermore, primary and secondary sources and data supporting the findings of this study were all publicly available at the time of submission.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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